Appendix A. Experiment Record for the Layer 2 Patient-State Model

This appendix records every experiment conducted between 5 September and 16 October 2026 to improve the HepatoTwin patient-state model, which predicts the controlled attenuation parameter (CAP; NHANES variable LUXCAPM, dB/m) measured by FibroScan transient elastography. For each experiment it states what was tested, the result, whether the change was adopted, and the reason. The cohort comprised participants with a complete elastography examination (LUAXSTAT = 1) and an observed CAP value, aged 12 to 80 years. All figures are taken from result files saved in the project repository.

A.1 Summary

Table A.1. Headline results.

Measure	Value	Context
Starting point	R^2 0.491	First XGBoost model, 14 inputs, fixed holdout (P1-test)
Best result on 2017–2020 data	R^2 0.524	128 inputs, pooled five-fold out-of-fold (P2)
Selected model, held-out test	R^2 0.493	Pooled 2017–2023 cohort, 3,060 test participants (P3-test)
Selected model, cross-validation	R^2 0.520	Training-set five-fold CV (P3-CV)
Steatosis classification	ROC-AUC 0.857 / 0.868	Calibrated classifiers for CAP ≥ 248 and ≥ 280 dB/m (P3-test)
Temporal validation	R^2 0.502; AUC 0.858 / 0.870	Trained on 2017–2020, tested on 2021–2023 (Section A.10)
Against clinical scores	AUC +0.018 vs FLI; +0.047 vs HSI	CAP ≥ 248, same 2,703 test participants (Section A.11)

Thirty-nine experiments varied the learning algorithm, the input features, the training weights, the treatment of the target and the training data. Every change of algorithm or feature set produced an R^2 within approximately 0.01 of the same plateau near 0.51. Adding new information helped modestly: approximately 80 additional NHANES variables increased R^2 by 0.010, and the 2021–2023 FibroScan cycle increased it by 0.004 on the original cohort.

The selected model pools both NHANES FibroScan cycles, uses 81 routinely collectable inputs, down-weights less reliable examinations, and is accompanied by isotonic-calibrated steatosis classifiers. It outperforms the original inputs trained on the same participants (test R^2 0.493 against 0.487) and covers every age group. The plateau is a property of the data: the

learning curve is flat, direct DXA measurement of visceral fat adds only 0.007, and even inputs taken from the FibroScan examination itself reach only 0.53.

A.2 Evaluation Protocols

Three evaluation protocols were used. Because each scores a different set of participants, R^2 values from different protocols are not directly comparable. Every result in this appendix is labelled with its protocol.

Table A.2. Evaluation protocols.

Code	Design	Use
P1	9,021 participants (2017–March 2020) split once 80/20 with seed 42 into 7,216 development and 1,805 test participants. Model choice by five-fold CV on the development set (P1-CV); test results reported as P1-test.	September and early October experiments. The test set was examined repeatedly during development, so P1-test is an internal comparison rather than an independent validation.
P2	All 9,021 participants, five-fold cross-validation with seed 42; R^2 computed on the pooled out-of-fold predictions. Baseline R^2 0.5135.	October model-family and new-data experiments.
P3	15,300 participants from both FibroScan cycles. A 20% test set (3,060, stratified by cycle, seed 42) set aside first; model choice by five-fold CV on the remaining 12,240 (P3-CV); test set scored once (P3-test).	Combined model and final selection.

Outcome categories used in the tables: **Adopted** (carried forward into the model), **Small gain** (improvement under about 0.01 R^2, not or not fully adopted), **No gain**, **Worse**, and **Diagnostic** (an analysis that explains model behaviour rather than a candidate model).

A.3 Phase 1: Baselines (5–6 September)

The initial models used 14 inputs: age, sex, race and ethnicity, BMI, waist circumference, ALT, AST, HbA1c, total cholesterol, HDL cholesterol, and day-one dietary energy, protein, carbohydrate and fat.

Table A.3. Phase 1 experiments: baselines.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E1	Baseline XGBoost	300 trees, depth 6, learning rate 0.05 on the 14 inputs.	P1-test R^2 0.491; RMSE 44.52 dB/m	Adopted	Established the reference. The early workflow imputed missing values before splitting, which leaks test information; corrected in E3.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E2	Algorithm comparison	XGBoost against linear regression and random forest.	P1-test R^2: XGBoost 0.491; linear 0.488; random forest 0.486	Small gain	XGBoost retained by a margin of 0.003. A linear model nearly matching a tree ensemble suggests the signal is largely additive and dominated by body-size variables. Selection here used test RMSE and was later replaced by CV-based selection.

A.4 Phase 2: Tuning, Additional Laboratory Inputs and the Supplied Model (to 18 September)

Table A.4. Phase 2 experiments: tuning and inputs.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E3	Leakage-free preprocessing and hyperparameter search	Imputation fitted within each fold; 40-configuration randomized search selected by CV RMSE.	P1-test R^2 0.485 (corrected baseline) → 0.506; CV RMSE 45.89 → 45.03; paired bootstrap 95% CI for RMSE gain 0.52–1.38 dB/m	Adopted	First statistically clear improvement. Shallow trees (depth 4), a low learning rate and row subsampling limit overfitting on a noisy target. Became the 14-input reference model.
E4	Second pass: 24 candidates, three input sets	14, 19 and 28 inputs (adding height, weight, GGT, uric acid, triglycerides, albumin, ALP, creatinine, bilirubin, total protein, globulin, iron, BUN, LDH); XGBoost variants, missing-value indicators, histogram boosting, spline ridge, an equal blend.	P1-CV R^2: 14 inputs 0.489; 19 inputs 0.496; 28 inputs 0.501; blend 0.502. P1-test 0.506 → 0.514; CI for RMSE gain −0.001 to 0.724	Small gain	Additional laboratory values added about 0.01, but the interval includes zero. Liver-relevant markers add little beyond waist and BMI, which already capture most of the metabolic signal.
E5	Reproduction of the supplied model (R^2 0.535)	Supplied 36-input script refitted on its own CAP-stratified split and on the original split.	Stratified split: supplied 0.535, project model 0.541. Original split: supplied 0.511, project model 0.514	Diagnostic	The reported 0.535 reflected an easier test split rather than a better model; on matched splits the project model scored higher. Single-split R^2 varies by roughly ±0.03 with test membership.
E6	Engineered features	36 raw inputs plus 10 derived features (waist-to-height, AST/ALT, TG/HDL, TC/HDL, partial HSI, FLI proxy, energy fractions, energy per kg) with missing-value indicators; 600 trees.	P1-CV R^2 0.506 (best of six); P1-test 0.515; CI for RMSE gain over E4 −0.30 to 0.38	Small gain	Became the 46-feature reference for October. Derived scores re-express inputs that the trees can already combine, so they contribute little new information.

A.5 Phase 3: Feature Ablations, Weighting and Regressor Choice (2 October)

All Phase 3 experiments used protocol P1, starting from the 36 raw inputs.

Table A.5. Phase 3 experiments: feature ablations and weighting.

ID	Experiment	What was tested	Result (P1-CV R^2)	Outcome	Interpretation
E7	Engineered-feature ablation	Raw only; plus four ratios; plus FLI and HSI; plus all ten engineered features.	0.5035 / 0.5048 / 0.5032 / 0.5036	Small gain	The four ratios added 0.001; the index scores added nothing. The ratios were kept.
E8	Individual ratios	WHtR, AST/ALT, TG/HDL and TC/HDL each added alone.	0.5022–0.5038 against 0.5035 (raw)	No gain	No single ratio carries information that the raw variables do not already provide.
E9	LDL cholesterol	46 features; plus LDL; plus LDL and a missing-value indicator.	0.5036 → 0.5020 / 0.5030	No gain	LDL is largely determined by total cholesterol, HDL and triglycerides, which are already inputs, and is missing for many participants.
E10	Recursive feature elimination	Feature counts from 46 down to 10.	46: 0.504; 25: 0.508; 15: 0.501; 12: 0.487; 10: 0.485	No gain	About 20–25 features carry nearly all of the signal. Waist, the FLI proxy, waist-to-height and BMI rank highest.
E11	Histogram gradient boosting	Same 40 features as XGBoost.	XGBoost 0.505; HistGradientBoosting 0.486	Worse	Default regularization fits this noisy target less well than the tuned XGBoost.
E12	Up-weighting extreme CAP values	Training weights increasing with distance from the mean, strength 0 to 0.75.	0.505 → 0.500 / 0.494 / 0.488	Worse	Prediction compression (E13) stems from missing information, not from the loss weighting. Weighting the extremes trades errors in the middle of the range for small gains at the tails.

A.6 Phase 4: Target Diagnostics and Model Families (13–16 October)

Diagnostics of the target variable were run first, as they explain why the subsequent model changes could not raise R^2 substantially.

Table A.6. Phase 4 diagnostics of the target and residuals.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E13	Residual analysis	Error and bias by CAP range on out-of-fold predictions (P2).	Slope of prediction on actual 0.51. Mean bias +44 dB/m for CAP < 200 and −67 dB/m for CAP > 390	Diagnostic	Regression toward the mean: the inputs cannot separate very lean or very fatty livers from average ones, so predictions shrink toward approximately 255 dB/m.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E14	CAP ceiling audit	Values at the device maximum and their examination quality.	129 of 9,021 (1.4%) exactly 400 dB/m; median CAP IQR 5 dB/m against 35 dB/m for other examinations	Diagnostic	CAP is truncated at 400 dB/m, so higher true values are censored. Too few cases to limit R^2 materially.
E15	Ceiling sensitivity	Model refitted and scored without CAP = 400 (P1-CV).	R^2 0.506 → 0.488; RMSE 44.28 → 43.74 dB/m	No gain	Errors fall slightly, but R^2 falls because removing the highest values reduces the variance against which R^2 is measured.
E16	Measurement quality by probe	Within-probe performance for M and XL probes (P1-CV).	Within-probe R^2: M 0.419, XL 0.278; RMSE 42.0 and 50.8 dB/m	Diagnostic	The XL probe is used for larger body habitus and yields noisier CAP; probe choice itself tracks obesity.

Table A.7. Phase 4 experiments: model families and target treatments.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E17	Probe-aware regression	Probe type (M or XL) as an input (P1-CV).	0.5062 → 0.5112	Small gain	Not used in the final model: probe type is recorded during the FibroScan examination, and a patient with a FibroScan already has a measured CAP.
E18	Probe weighting	XL examinations weighted 1.5× and 2×.	0.5112 → 0.5080 / 0.5018	Worse	XL examinations are noisier; giving them more weight increases the noise learned.
E19	CatBoost	Same features and folds as XGBoost (P2).	XGBoost 0.5135; CatBoost 0.5093	No gain	A different boosting implementation on the same information reaches the same plateau.
E20	Classification-to-regression hybrid	Binary or four-class steatosis probabilities supplied to the regressor, nested CV (P2).	0.5135 → 0.5116 / 0.5129	No gain	Class probabilities are functions of the same inputs and add no new information.
E21	Linear recalibration	Cross-fitted linear correction to stretch compressed predictions (P2).	0.5135 → 0.5127	No gain	Squared-error models are already calibrated in the mean; stretching predictions widens errors as much as it removes bias.
E22	Quantile regression	Median (q50) model and an 80% interval from q10 and q90 (P1-CV).	q50 R^2 0.503 against 0.506; interval coverage 74.7% (nominal 80%); mean width 98 dB/m	No gain	Useful for expressing uncertainty, not for accuracy. The interval width indicates that an individual CAP is uncertain by roughly ±50 dB/m.
E23	Fold-local feature selection	Lasso and ExtraTrees RFE fitted within each fold (P1-CV).	0.506 → Lasso 0.499; RFE 0.504	No gain	Consistent with E10: weak features are not harming the model.
E24	Alternative learners	Extra Trees, random forest and ridge regression on the 46 selected features (P1-CV).	XGBoost 0.506; Extra Trees 0.500; random forest 0.498; ridge 0.482	No gain	Every model family reaches the same ceiling.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E25	Neural network	Multilayer perceptron with layers (128, 64) (P1-CV).	0.453	Worse	Tabular data of this size with substantial missingness favours tree ensembles.
E26	Stacking	XGBoost, Extra Trees and random forest with a ridge meta-learner (P1-CV).	0.506 → 0.509	Small gain	A gain of 0.003 for three times the training cost and complexity; not adopted.
E27	Steatosis classification	Binary (S0 against S1–S3) and four-class; provisional cut-offs 248, 268 and 280 dB/m (P2).	Binary ROC-AUC 0.862, balanced accuracy 0.776. Four-class macro F1 0.439, one-vs-rest AUC 0.742	Adopted	The binary task matches the evaluation specified in the proposal (AUC, F1, calibration) and performs well. The four-class task fails for S1 and S2, which span only 20 and 12 dB/m, narrower than the model error.
E28	HbA1c ablation	Model with and without HbA1c on the same participants.	Not completed	—	The script stopped because an expected parameter file was never created; no results are available.

A.7 Phase 5: Additional Data (16 October)

Phase 4 indicated that the limit lay in the available information rather than in the model, so Phase 5 added NHANES components not previously used. Elastography fields and self-reported liver conditions were excluded as inputs throughout, because they would act as proxies for the target.

Table A.8. Phase 5 experiments: additional data.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E29	Extended NHANES variables	Added in cumulative groups (P2): body measures; fasting insulin and glucose (HOMA-IR, TyG); blood count, CRP and ferritin; blood pressure; questionnaires (diabetes, hypertension, weight history, alcohol, smoking, activity, sleep); prescriptions.	0.5135 → body 0.5131 → fasting 0.5175 → blood count 0.5173 → blood pressure 0.5171 → questionnaire 0.5232 → prescriptions 0.5221; tuned, 128 inputs 0.5239	Small gain	Largest single gain in the record (+0.010). Insulin resistance and diabetes or weight history add information not fully captured by body size. Prescriptions and several questions exist only in 2017–2020.
E30	Learning curve	Training on 25%, 50%, 75% and 100% of each training fold (P2).	0.488; 0.501; 0.511; 0.512	Diagnostic	The curve flattens: the final 25% of data adds 0.002, so additional participants alone would help only slightly.

ID	Experiment	What was tested	Result	Outcome	Interpretation
E31	2021–2023 FibroScan cycle	6,279 additional participants with the same examination and variable names, added to training only (P2 folds).	0.5135 → 0.5170 on the 2017–2020 participants	Small gain	Consistent with the learning curve. The 2021–2023 cycle is the final NHANES cycle with liver elastography.
E32	DXA body composition	Visceral, subcutaneous, android and gynoid fat (2017–2018, ages 8–59; 3,307 linked participants).	0.5094 → 0.5168 on the DXA subset	Small gain	Visceral fat is the strongest known correlate of liver fat, yet it adds only 0.007. Unsuitable for the final model: available for part of one cycle and not a routine test.
E33	Measurement-noise estimate	Standard error of each examination's median CAP from its IQR and number of valid measurements.	Within-examination noise ≈ 3.7% of CAP variance	Diagnostic	Noise within a single examination does not explain the ceiling; between-visit variation cannot be measured in NHANES.
E34	Upper-bound check with examination-derived inputs	Liver stiffness and probe type from the same FibroScan examination (pooled cohort, five-fold CV).	0.518 → with probe 0.524; with stiffness 0.529	Diagnostic	These are not legitimate predictors, since they come from the measurement being predicted, and they still remain near 0.53.
E35	Combined model	Both cycles (15,300 participants); 36 core inputs plus 45 variables collected identically in both cycles; training weighted by examination reliability (P3).	P3-test R^2: 2017–2020-only reference 0.480; same inputs pooled 0.487; combined 0.493. Reliable examinations 0.517 → 0.533. P3-CV 0.520	Adopted	Best full-population result. This test split is harder than average (the reference scores 0.480 on it against about 0.51 in CV), so the difference between models is the meaningful quantity.
E36	Cohort filters	Each filter trained and scored within the filtered cohort; mean of three CV seeds.	All 0.518; CAP IQR < 40 0.548; ages 12–59 0.548; IQR < 40 and ages 12–59 0.584; adults only 0.498; without CAP 400 0.497; IQR < 40 and ages 60+ 0.414	Diagnostic	The highest value in the record (0.584) is stable across seeds and age cut-offs of 50 to 65 years, but excludes 59% of participants, including everyone aged 60 or over. Not adopted.
E37	Performance by age group	Error by age group on pooled out-of-fold predictions.	R^2: 12–17 0.43; 18–39 0.54; 40–59 0.49; 60+ 0.41. RMSE 40.5, 42.2, 44.1, 46.0 dB/m	Diagnostic	Error in the 60+ group is only slightly larger; R^2 is low mainly because CAP varies less within that group (SD 59.6 against 63.0 dB/m overall).
E38	Improving the 60+ group	2017–2020 history and medication items; a 60+-only model; doubled weighting; deeper trees.	60+ R^2 0.405 → 0.406 / 0.394 / 0.406 / 0.408	No gain	Determinants of liver fat in older adults, such as genetic risk, accurate alcohol history, muscle loss and other liver disease, are not measured in NHANES 2017–2023.
E39	Calibrated steatosis classifiers	XGBoost classifiers for CAP ≥ 248 and ≥ 280 dB/m with isotonic calibration on training out-of-fold predictions and an F1-selected threshold (P3-test).	CAP ≥ 248: AUC 0.856, Brier 0.154, calibration error 0.025 → 0.018. CAP ≥ 280: AUC 0.868, calibration error 0.024 → 0.019	Adopted	Additional inputs leave discrimination unchanged but reduce calibration error by about a quarter. Calibrated probabilities are required by the scenario engine.

E30, E33 and E34 were exploratory runs whose outputs were not saved to the repository. During Phase 5, questionnaire cleaning initially treated the codes 7 and 9 as missing for every item, which also removed valid answers such as 7 hours of sleep or 9 drinks. Missing-value codes were made item-specific and all affected experiments were rerun; R^2 changed by less than 0.001. The figures reported are from the corrected runs.

A.8 Final Model Selection

The selected model is an XGBoost regressor (1,500 trees, depth 4, learning rate 0.01, column subsampling 0.4) trained on 12,240 participants from both FibroScan cycles. It uses 81 raw inputs and six derived scores: waist-to-hip ratio, HOMA-IR, the triglyceride-glucose (TyG) index, mean systolic and diastolic blood pressure, and one-year weight change. Examinations with a CAP IQR of 40 dB/m or more are down-weighted (weight 40/IQR, minimum 0.25) rather than excluded. Two isotonic-calibrated classifiers estimate P(CAP ≥ 248) and P(CAP ≥ 280). The hyperparameters and examination weighting were chosen by training-set cross-validation alone.

Table A.9. Selected model, held-out test performance (P3-test, scored once). The reference uses the 36 core inputs with the same training participants and settings.

Test group	n	Selected R^2	Reference R^2	Selected RMSE (dB/m)
All participants	3,060	0.493	0.487	43.6
Reliable examinations (CAP IQR < 40)	1,891	0.532	0.522	42.2
Ages 12–17	400	0.376	0.366	40.0
Ages 18–39	868	0.456	0.461	44.5
Ages 40–59	772	0.490	0.486	43.7
Ages 60+	1,020	0.418	0.400	44.1

Table A.10. Calibrated steatosis classifiers, held-out test performance (P3-test). Cut-offs are provisional.

Target	ROC-AUC	Brier	Calibration error	F1	AUC, ages 60+
CAP ≥ 248 dB/m (any steatosis)	0.857	0.154	0.023	0.792	0.823
CAP ≥ 280 dB/m	0.868	0.137	0.020	0.704	0.844

Reasons for the selection

1. **Best full-population result.** The model improves on the same inputs trained on the same participants, overall and in the 60+ group, and was chosen by cross-validation rather than by the test set.
2. **No exclusions.** It covers ages 12 to 80 and all examination qualities, as required of a patient-state model for the digital twin.

3. **Routinely collectable inputs only.** Body measurements, standard blood tests, blood pressure and a short questionnaire. No input comes from the FibroScan examination or from DXA.
4. **Alignment with the proposal.** The proposal's evaluation plan (Section 12.1) specifies ROC-AUC, F1, Brier score and calibration; the calibrated classifiers report all four.
5. **Tolerance of missing data.** Fasting laboratory values are missing for about half of participants; the model accepts missing inputs.

Alternatives not selected

Table A.11. Alternatives considered and reasons for rejection.

Alternative	Result	Reason not selected
Reliable examinations, ages 12–59 only	R^2 0.584 (CV)	Excludes 59% of participants, including everyone aged 60 or over.
128-input model	R^2 0.524 (P2)	About 30 of its variables were not collected in 2021–2023; adding them to the pooled cohort raised CV R^2 by only 0.002 (0.518 to 0.519).
Probe type or liver stiffness as inputs	R^2 0.524–0.529	Both are produced by the FibroScan examination and are unavailable without a measured CAP.
Stacking, CatBoost, MLP, hybrid models	R^2 0.453–0.513	No gain, or a gain too small to justify the added complexity.
DXA visceral fat	+0.007 R^2	Available for one partial cycle only and not a routine test.

A.9 Why R^2 Plateaus Near 0.5

Seven independent lines of evidence indicate that the remaining variation in CAP is not explained by routine clinical measurements:

1. Every algorithm tested (linear regression, random forest, Extra Trees, histogram boosting, XGBoost, CatBoost, a neural network and stacking) reaches an R^2 between 0.45 and 0.52 on the same inputs (E2, E11, E19, E24–E26).
2. Feature engineering, selection and ablation change R^2 by at most 0.003 (E7–E10, E23).
3. The learning curve is flat; 6,279 additional participants add 0.004 (E30, E31).
4. Direct DXA measurement of visceral fat adds 0.007 (E32).
5. Approximately 80 additional NHANES variables add 0.010 (E29).
6. Even inputs taken from the FibroScan examination itself reach only 0.53 (E34).

7. Within-examination measurement noise is about 4% of the variance, so the unexplained variance reflects biological differences between people with similar measured profiles, such as genetic risk, alcohol intake and dietary history (E33).

Accordingly, the full-population R^2 (0.493 on the held-out test set; 0.520 in cross-validation) is reported together with ROC-AUC and calibration, and the plateau is presented as a finding. Results for restricted cohorts, such as reliable examinations only, are reported alongside the full-population figure and never in place of it.

A.10 Additional Validation of the Selected Model

After selection, five further analyses (V1–V5) tested the selected model against the limitations identified during development. None of them changed the model or its settings, and the test set was not used to choose anything. Intervals are 95% percentile bootstrap intervals from 1,000 participant resamples of the test set.

Table A.12. Additional validation analyses.

ID	Experiment	What was tested	Result	Outcome	Interpretation
V1	Uncertainty	Bootstrap intervals for every test metric, overall and by age group.	R^2 0.493 (0.461–0.521); AUC 0.857 (0.844–0.869) and 0.868 (0.855–0.881).	Reported	The intervals are narrow; the classifiers are clearly in the 0.84–0.88 AUC range.
V2	Survey weighting	Test metrics weighted by NHANES examination weights (WTMECPRP, WTMEC2YR), pooled in proportion to cycle length.	R^2 0.507; AUC 0.863 and 0.875; calibration error 0.027 and 0.022.	Reported	Performance holds for the weighted US population and is slightly higher than unweighted.
V3	Older adults	Separate isotonic calibration for each age group, fitted on the same training out-of-fold predictions, compared with the single global calibration.	Ages 60+ calibration error 0.037→0.041 (CAP ≥ 248) and 0.050→0.047 (CAP ≥ 280); overall slightly worse.	Not adopted	No consistent gain. The 60+ test calibration error lies within its interval (0.030–0.079), and is 0.030 and 0.029 on the larger temporal test (n = 2,352).
V4	Reference standard	Third classifier at the stricter cut-off CAP ≥ 302 dB/m, trained and calibrated in the same way.	AUC 0.872 (0.857–0.884); Brier 0.116; calibration error 0.016.	Reported	Discrimination is stable across three definitions of steatosis.
V5	Temporal validation	Training on all 9,021 participants from 2017–March 2020 with the selected settings; testing on all 6,279 participants from 2021–2023.	R^2 0.502 (0.482–0.519); AUC 0.858 and 0.870; calibration error 0.023 and 0.019.	Reported	No loss of performance on participants examined two to six years after the training data.

Table A.13. Selected model with 95% bootstrap intervals (P3-test, n = 3,060) and temporal validation (trained on 2017–2020, tested on 2021–2023, n = 6,279).

Measure	P3-test, unweighted	P3-test, survey-weighted	Temporal test
CAP R^2	0.493 (0.461–0.521)	0.507 (0.464–0.544)	0.502 (0.482–0.519)
CAP RMSE (dB/m)	43.6 (42.3–45.0)	44.0 (41.9–46.1)	44.6 (43.6–45.5)
AUC, CAP ≥ 248	0.857 (0.844–0.869)	0.863 (0.844–0.881)	0.858 (0.848–0.867)
AUC, CAP ≥ 280	0.868 (0.855–0.881)	0.875 (0.856–0.893)	0.870 (0.861–0.879)
AUC, CAP ≥ 302	0.872 (0.857–0.884)	0.874 (0.853–0.894)	–
Brier, CAP ≥ 248	0.154 (0.147–0.161)	0.150 (0.140–0.161)	0.153 (0.148–0.159)
Brier, CAP ≥ 280	0.137 (0.130–0.144)	0.135 (0.125–0.145)	0.136 (0.131–0.141)
Calibration error, CAP ≥ 248	0.023 (0.018–0.039)	0.027 (0.021–0.052)	0.023 (0.017–0.033)
Calibration error, CAP ≥ 280	0.020 (0.015–0.037)	0.022 (0.017–0.047)	0.019 (0.014–0.030)
AUC ages 60+, CAP ≥ 248 / 280	0.823 / 0.844	–	0.811 / 0.832

Calibration error is the expected calibration error over ten equal-sized bins. It is biased upward in small samples, which is why its intervals are asymmetric and why the 60+ estimate is lower on the larger temporal test set. Survey strata and primary sampling units were not used, so the intervals reflect participant sampling rather than the full survey design.

A.11 Comparison with Clinical Scores and Figures

The selected classifiers were compared with two established non-invasive steatosis scores on the same held-out participants: the Fatty Liver Index (FLI; Bedogni et al., 2006), computed from BMI, waist circumference, triglycerides and GGT, and the Hepatic Steatosis Index (HSI; Lee et al., 2010), computed from the ALT/AST ratio, BMI, sex and diabetes. The comparison used the 2,703 test participants with every input to both scores. Differences in AUC are paired, with 95% intervals from 1,000 bootstrap resamples.

Table A.14. Discrimination of the selected classifiers and two clinical scores on the same test participants (n = 2,703). Δ AUC is the model minus the score.

Target	Model AUC	FLI AUC	Δ AUC vs FLI	HSI AUC	Δ AUC vs HSI
CAP ≥ 248 dB/m	0.860	0.842	+0.018 (0.012–0.025)	0.813	+0.047 (0.038–0.057)
CAP ≥ 280 dB/m	0.870	0.843	+0.028 (0.021–0.035)	0.817	+0.053 (0.044–0.064)
CAP ≥ 302 dB/m	0.874	0.841	+0.033 (0.025–0.042)	0.812	+0.062 (0.049–0.074)

The model discriminates better than both scores at every cut-off, and every interval excludes zero. The margin over FLI is modest, which is expected: FLI is itself a strong predictor and a derived FLI value is one of the model inputs. The gain grows with stricter cut-offs. FLI was computed with non-fasting serum triglycerides, because fasting values exist for only part of the cohort, which may slightly disadvantage it.

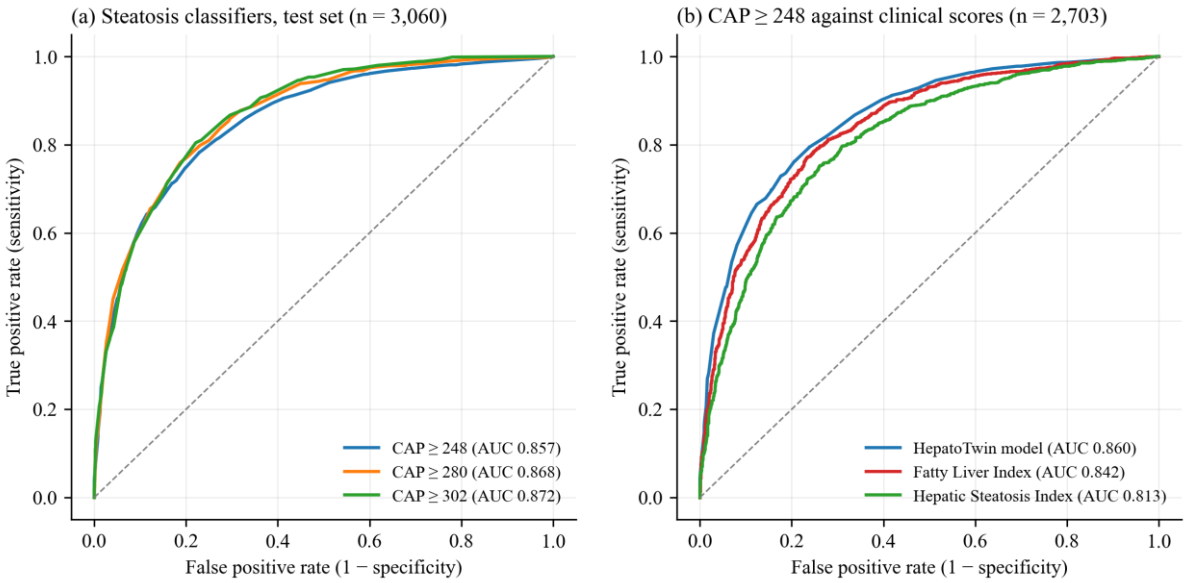


Figure A.1. Receiver operating characteristic curves on the held-out test set. (a) The three steatosis classifiers. (b) The CAP ≥ 248 dB/m classifier against the Fatty Liver Index and the Hepatic Steatosis Index on the 2,703 participants with complete inputs.

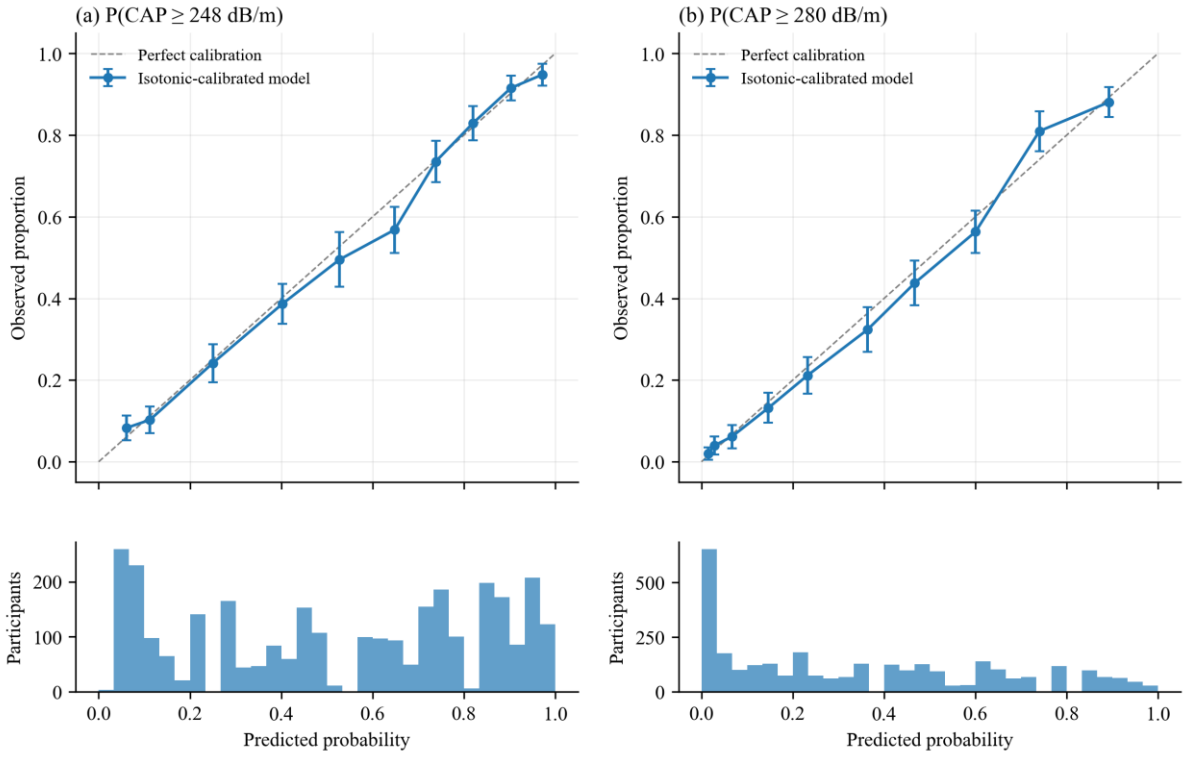


Figure A.2. Calibration of the isotonic-calibrated classifiers on the held-out test set (n = 3,060). Points are deciles of predicted probability with 95% binomial intervals; the histograms show the distribution of predicted probabilities.

The SHAP summary (Figure A.3) shows that the classifier relies mainly on central adiposity (waist circumference and waist ratios), followed by body weight, age, HbA1c, creatinine and

the triglyceride-glucose index. SHAP values were computed with XGBoost's exact TreeSHAP on the uncalibrated model; isotonic calibration is monotone, so the ranking of features is unchanged. SHAP describes how the model uses its inputs and does not establish causal effects.

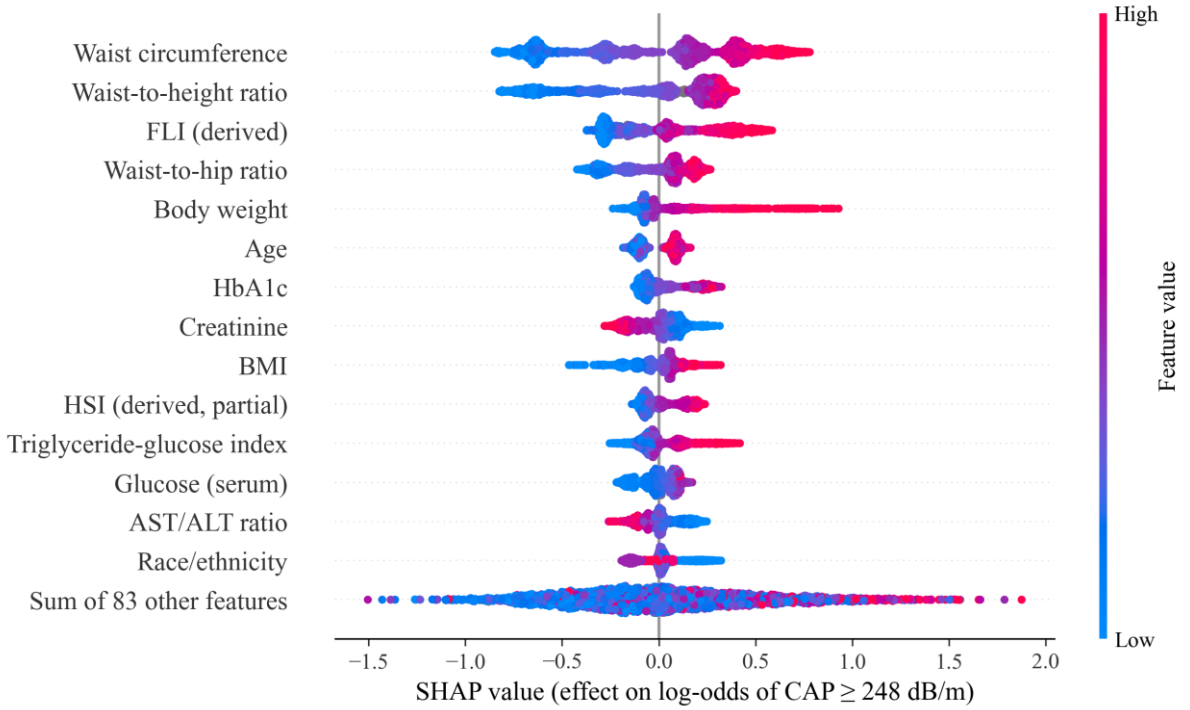


Figure A.3. SHAP summary for the CAP ≥ 248 dB/m classifier on the held-out test set. Each point is one participant; colour shows the input value and position its effect on the log-odds of steatosis.

The learning curve (Figure A.4) refits the selected settings on random subsets of the training set. Test R^2 rises from 0.441 with 1,224 participants to 0.493 with 9,180 and does not improve with the full 12,240; AUC follows the same pattern (0.839 to 0.856). This confirms that the plateau described in Section A.9 is not caused by a shortage of participants.

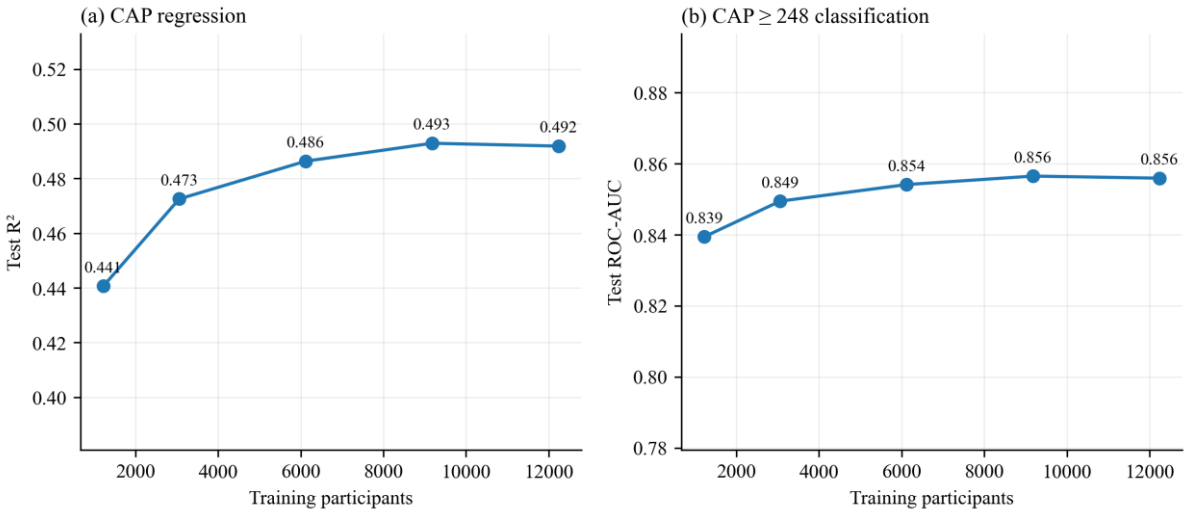


Figure A.4. Learning curves for the selected settings: held-out test performance against the number of training participants.

A.12 Limitations

- All validation is internal to NHANES. The temporal validation (V5) uses later participants from the same survey programme, not an external population. The P1 test set was reused during development, and every reported figure carries some selection optimism.
- Survey weights were applied in evaluation (V2) but not in training, and survey strata and clustering were not used for variance estimation.
- CAP is an imperfect measure of liver fat and is truncated at 400 dB/m. No biopsy reference is available in NHANES; results were therefore checked at three cut-offs (V4).
- The steatosis cut-offs (248, 280 and 302 dB/m) follow published CAP thresholds but have not been verified against histology in this cohort.
- The data are cross-sectional; the model describes current state and cannot predict response to dietary change.
- Single-split test R^2 varies by about ±0.03 with test membership (E5), so differences below 0.01 between models are within noise.